

Libor Reform

Impact on Derivatives Portfolio

Julien Second – Royal Bank of Canada

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“Libor is measuring the rate at which banks are not borrowing from one another”

Andrew Bailey, CEO of the FCA

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- Main users of Libor
 - Loans
 - FRNs
 - Derivatives
- A total of \$350 trillion of financial contracts is affected
- End 2021, panel banks will not have to publish Libor anymore
- ISDA is working on a fallback solution based around ARR* + static historical spread
- Market participants may decide to follow this fallback... or not
- Even if they do, significant challenges to overcome

* Alternative Reference Rate

- Fallback defines a mechanism to calculate cash flows ...
- ... But not a complete valuation mechanism, which is
 - Forecast cash flows
 - Discount cash flows
 - Take into account contract specifics
- Multi-curve frameworks need to be defined
- Liquidity in AAR instruments needs to build up to be able to construct curves
- Very likely that Libor might co-exist for a period of time with ARR, post 2021

- Most Libor-based contracts are term-forward looking
- Main differences between ARR and Libor
 - ARR **overnight** rate vs Libor **term** rate
 - ARR **backward** looking vs Libor **forward** looking
 - ARR **secured or unsecured** vs Libor **unsecured** (extra credit component)
- Need to come up with a mechanism that converts ARR overnight backward looking to ARR term forward looking
- Mechanism will be model-dependent
 - payment lags: convexity adjustment
 - volatilities of ARR (either estimated or market implied, if liquidity builds up)
- Being model-dependent, valuations will be more open to disagreements

- **FRA**
 - Building block of other Libor-type instruments, such as swaps
 - Pays a rate which is o/n compounding as opposed to term

- **Caps/Floors**
 - Options on FRAs: becomes an Asian option (average)
 - Need a volatility input (estimated or implied if market develops)
 - Potentially changing the strike of existing options

- **Cross-Currency Swaps**
 - Similar constraints to FRAs
 - Libor reform not coordinated between benchmark of different currencies
 - Potentially affecting each leg differently

- **Swaptions/Callables**
 - Potential change in the strike of the swaption or callable (exercise boundary)
 - Impact on the average duration of the callable

- **XVA**
 - Exposures, function of forwards and volatilities (both impacted)
 - Banks still marking their FVA (unsecured discounting) to Libor as a proxy for their bank spread, will need to come up with a mark properly reflective of their bank spread

- Model dependency introduced in valuations
- However as liquidity in certain ARR-based products develops, this dependency can be minimised
- Why and how? (Loose) analogy with FRA / Futures
 - Knowing the price of a futures, we can use a model to imply the price of a FRA (model dependent)
 - From there we can imply the price of a swap as a string of FRAs
 - However if a FRA market exists, we can use it **directly** as an input for a swap curve construction
 - Why? Because the features of a FRA more naturally aligns with these of a swap => Less model dependency than using a futures model
- Since the move from Libor discounting to OIS discounting, derivatives markets are already used to deal with certain ARR products
- Volumes are still building up in ARR contracts, although the pace is irregular
- Derivatives market expected to lead the way in the transition
- A few recent new FRN issues in SOFR and SONIA. More needed to support the derivatives market.

Current state of liquidity [1]

■ USD SOFR

- Monthly swap volumes ~ 3 bn
- Small compared to the existing RFR (Fed Funds) monthly volumes ~ 7 trillion
- Gathering pace but long way to go

■ GBP SONIA

- Monthly swap volumes ~ 2.5 trillion
- Many LDI* funds have already switched from Libor to Sonia.
- Good two way flows between asset swappers (receiver of Sonia) and LDI funds (payer of Sonia)
- Widening of the Sonia/Libor basis about six months ago, liquidity built up

■ EUR ESTER

- Ester will be first published in October 2019
- For reference, monthly swap volumes in the existing RFR (EONIA) ~ 3.5 trillion

■ AUD AONIA

- Monthly swap volumes ~ 1.2 trillion, mostly below 2y

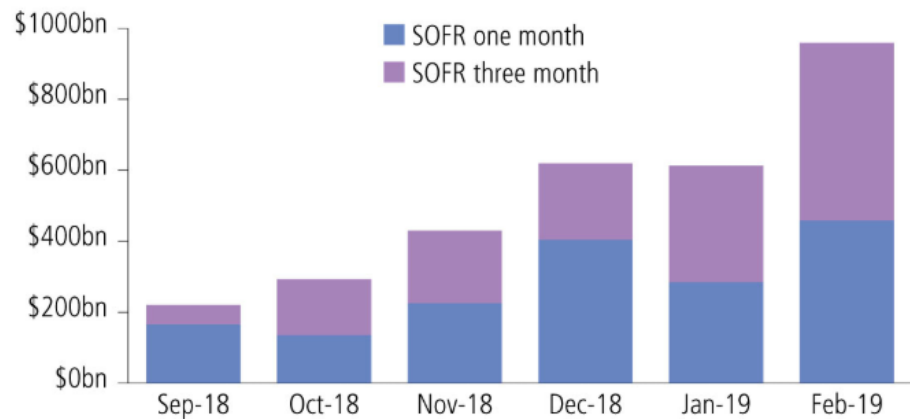
■ CHF SARON

- Monthly swap volume ~ 320 bn (approx. 10% of the CHF Libor market)

* Liability-Driven Investment

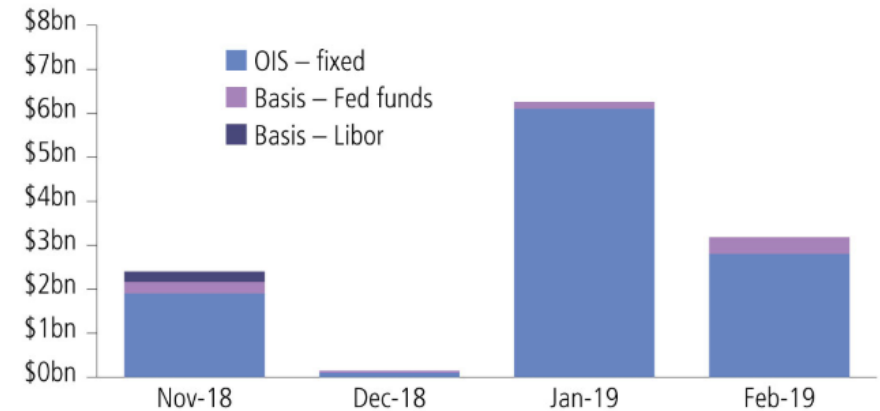
Current state of liquidity [2]

1. CME SOFR futures



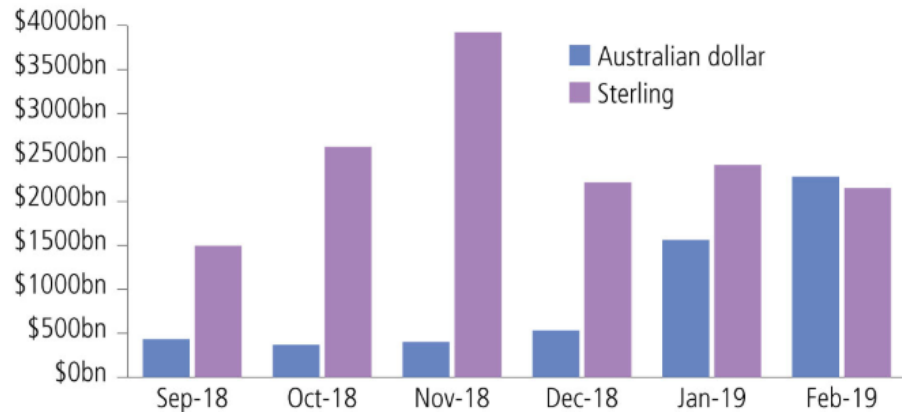
Source: ClarusFT

2. SOFR swaps



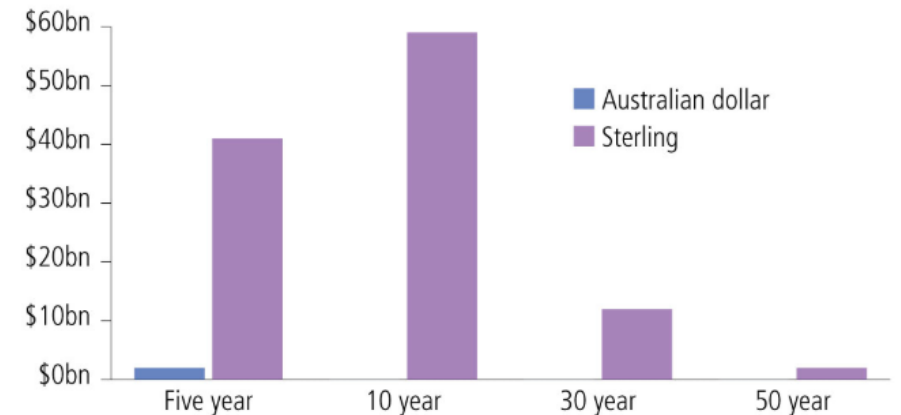
Source: ClarusFT

3. LCH SwapClear overnight indexed swaps volume



Source: ClarusFT

4. LCH SwapClear overnight indexed swaps by tenor (Feb 2019)



Source: ClarusFT

Source: Risk Magazine

- Identify Libor exposure per counterparty (see next two slides)
- Order by size of exposure and counterparty type
 - Large derivatives users (Interbank, ..)
 - Institutionals
 - Other financial institutions
 - Corporates
 - Other users
- Engage discussions with counterparties as early as possible
- Potential differences of valuations as a new XVA?
 - At this stage: unlikely
 - However bid/offer reserves around Libor exposures can be considered

- The Libor exposure is the change of PV when adding 1bp to a Libor leg. Also known as PV01 or forecasting risk.
- It is different from the DV01 risk (sensitivity to a rates curves shift)
- At a counterparty level, the Libor exposure gives a sense of how much value is potentially open to disagreement when Libor retires
- When running risks on derivatives portfolios, we usually get
 - **DV01s**: change of PV due to 1bp shift in a given rates curve (swaps, xccy, basis..)
 - **Discounting risk**: change of PV due to 1bp shift in discounting curve
- $DV01 = \text{forecasting risk} + \text{discounting risk}$
- We can use this to approximate the forecasting risk (i.e Libor exposure)

Libor exposure: example

- 500m USD swap 10y, receive 5% vs pay USD Libor 3m, discounted at Eonia (bilateral)
 - PV = +97.9m
 - USD Swap DV01 = -484k
 - EUR/USD Xccy basis DV01 = +50k
 - Euribor/Eonia basis DV01 = +50k
 - Discounting DV01 = -50k

- Removing the discounting risk from the respective DV01s (applying the correct conventions/signs) we can isolate the forecasting risk
 - USD swap = -434k (i.e risk to USD Libor 3m)
 - Other: 0, as the basis risk was purely a discounting effect

- The exact Libor exposure (PV01) is -436k vs the approximation of -434k

- This approximation will quantify the Libor exposure of all Libor-type instruments (swaps, xccy, basis, etc..) facing a given counterparty

- This can be extended to the entire derivatives portfolio

- LCH announced that in 2020 SOFR will replace USD OIS (Fed Funds) as both the rate paid on collateral and the rate used for discounting
- This will apply to both existing and new USD swaps
- Discounting change exercise: similar to the move from Libor to OIS discounting in 2008
- Change of value for existing swaps: compensation mechanism being studied

- When LCH moves to SOFR for discounting, what will happen?
- 500m USD payer swap 10y bilateral (Fed Funds) hedged with the opposite cleared swap (SOFR)
 - If USD swap rates move by 1%, the package picks up \$20k 01 of SOFR/Fed Funds basis DV01
 - If SOFR/Fed Funds basis move by 10bp, the package picks up \$5K 01 of USD swap DV01
- A discounting mismatch between client trades still at Fed Funds and their hedges at SOFR will lead to basis risks that need to be dynamically managed
- This dynamics which already exists is called Cross-Gamma, which at a large scale is difficult to manage
 - Risks highly multidimensional (driven by Rates, Inflation, FX, ...) and volatile
 - Dynamic hedging sometimes “crowded”, as dealers may have similar positions
 - Liquidity on long dates limited

- Since 2008, in an effort to mitigate Cross-Gamma effects, collateral agreements have been negotiated with counterparties...
- ... so that their rate align with these of the CCPs, used for hedging
 - With USD-heavy counterparties, hedged mainly facing CCP at Fed Funds, collateral agreements would be at Fed Funds
 - With EUR-heavy counterparties, hedged mainly facing CCP at Eonia, collateral agreements would be at Eonia
 - Etc..
- If CCPs do switch to AARs, it is also likely that in time collateral agreements will also be re-negotiated from OIS (current) to AARs